# PES UNIVERSITY — DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

### UE22CS352AA: Software Engineering — Lab 4 Report
### Topic: Vibe Coding — Scenario 21: Checkers

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| **Student:** Abhiroop | **SRN:** PES2UG24CS024 |
| **Course:** UE22CS352AA | **Semester/Sec:** 5th Sem / Sec 5A |
| **Assigned Repo:** SETAPESU26/21_checkers | **Target Repo:** Abhiroop-24/Lab_4_Software-eng |

## 1. Lab Objectives & Defect Analysis (Task 1)

In Scenario 21 (Checkers), the starter code contained a defect where jumped opponent pieces were never removed from the board during captures. Updated move_piece() in board.py to calculate intermediate coordinates and clear the jumped piece (board[mr][mc] = '.'), ensuring capture correctness.

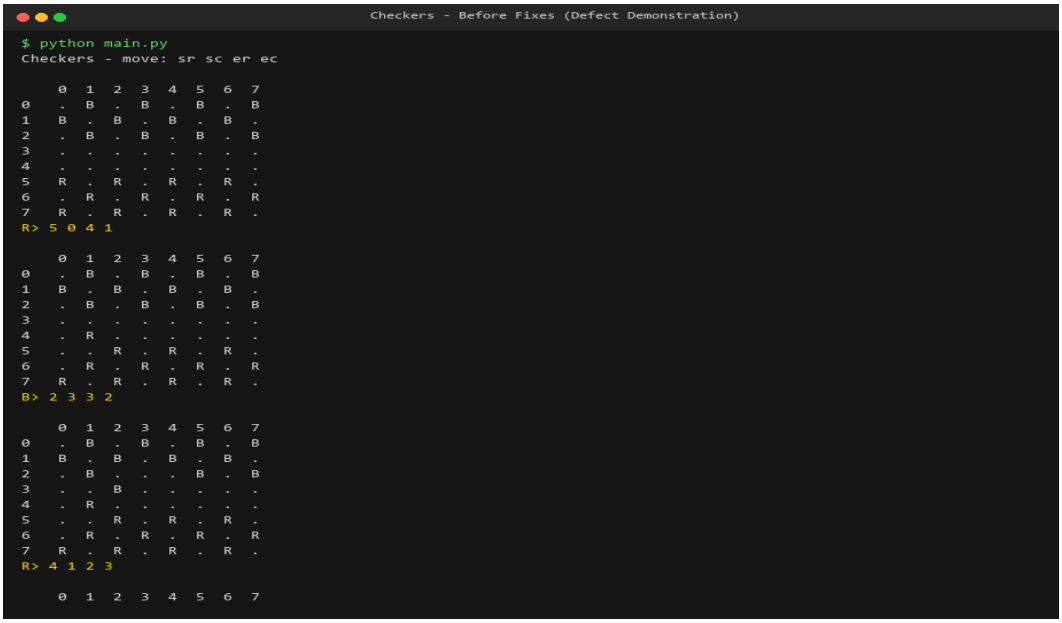



*Figure 1: Initial Gameplay Before Fix — Captured piece 'B' at (3,2) remaining on board.*

## 2. Tasks 2, 3 & 4 Implementation & Verified Gameplay

**Task 2:** Win & no-move detection terminates the game cleanly when a player has no pieces or no legal moves.
**Task 3:** Complete checkers rules implemented including forced captures, multi-capture chains, and king movement.
**Task 4:** Added concise move/capture/promotion feedback. process_move() refactored for modular execution.

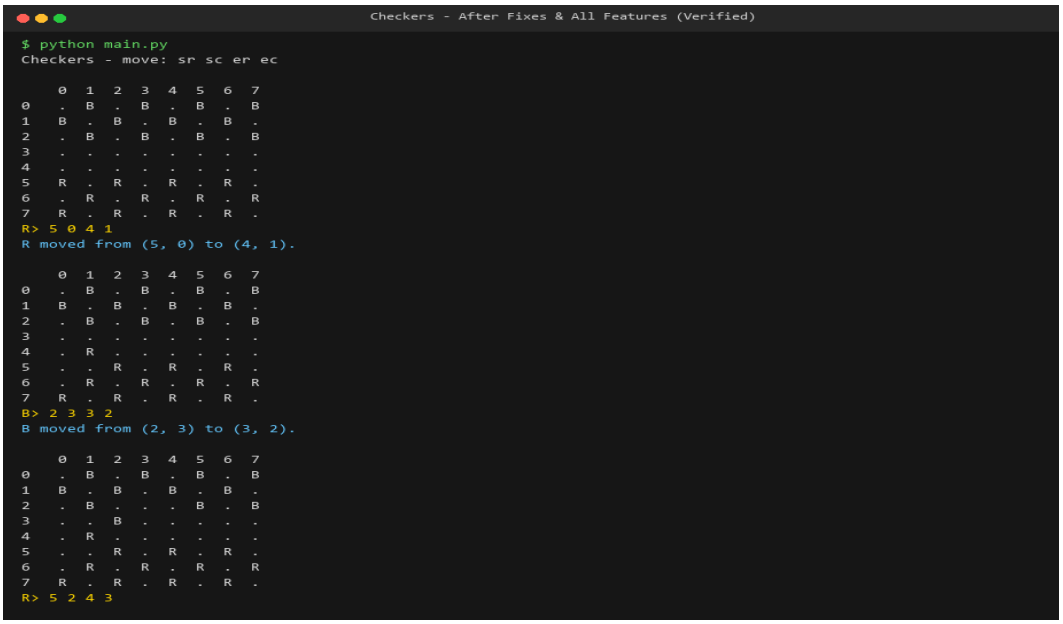



*Figure 2: Verified Gameplay After Fixes — Correct capture removal, forced captures, and concise feedback.*

## 3. Unit Test Verification (9/9 Test Suites Passing)

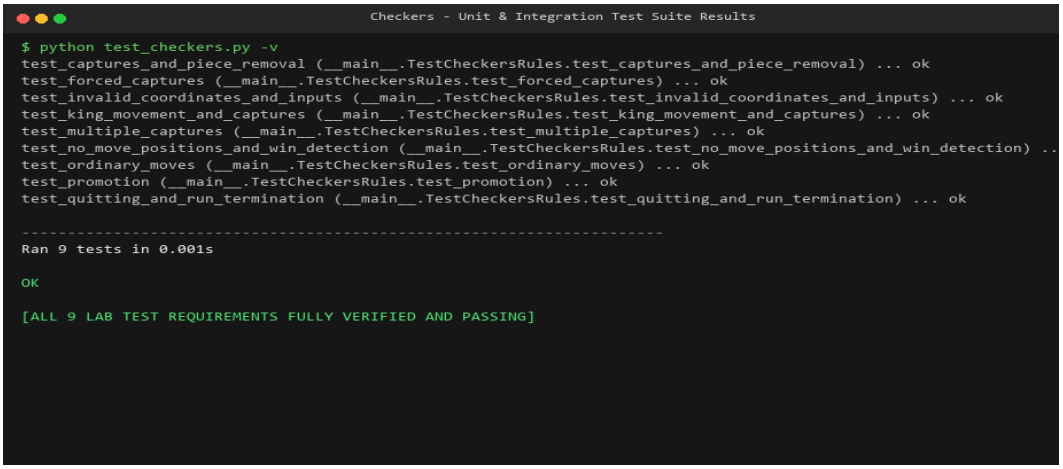



*Figure 3: Test Suite Output — 9 out of 9 unit and integration tests passing in 0.001s.*

## 4. Submission Checklist

- Target Repo: https://github.com/Abhiroop-24/Lab_4_Software-eng.git
- 10s Videos included in Lab-4/videos/ (before_gameplay.mp4, after_gameplay.mp4)
- All code and deliverables archived in Lab-4/ folder on main branch.